

Exploring transformer models in the sentiment analysis task for the under-resource Bengali language

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ABSTRACT

In the sentiment analysis (SA) task, we can obtain a positive or negative-typed comment or feedback from an online user or a customer about any object, such as a movie, drama, food, and others. This user's sentiment may positively impact various decision-making processes. In this regard, a lot of studies have been done on identifying sentiments from a text in high-resource languages like English. However, a small number of studies are detected in the under-resource Bengali language because of the unavailability of the benchmark corpus, limitations of text processing application software, and so on. Furthermore, there is still enough space to enhance the classification performance of the SA task. In this research, we experiment on a recognized Bengali dataset of 11,807 comments to find positive or negative sentiments. We employ five state-of-the-art transformer-based pretrained models, such as multilingual Bidirectional Encoder Representations from Transformers (mBERT), BanglaBERT, Bangla-Bert-Base, DistilmBERT, and XLM-RoBERTa-base (XLM-R-base), with tuning of the hyperparameters. After that, we propose a combined model named Transformer-ensemble that presents outstanding detection performance with an accuracy of 95.97% and an F1-score of 95.96% compared to the existing recent methods in the Bengali SA task.

1. Introduction

Social media, online reviews, news articles, and various online sources generate enormous quantities of textual data in the digital age today. Extracting meaningful insights from this unstructured data has become a formidable obstacle. Identifying the sentiment or emotional tone as positive, negative, or neutral from this insight is known as sentiment analysis (SA). SA tasks enable individuals and organizations to obtain an awareness of public opinion and emotional responses. In the past, organizations apply to spent a lot of time on this task and frequently found it difficult to assess market demand and recent changes. At present, SA is widely utilized in many contexts. Manufacturers can elevate and refine their product or service content by gathering user feedback, whether positive or negative, using opinion mining (Ireland and Liu, 2018). Again, sentiments, ideas, and emotions among individuals play a significant role in decision-making. While purchasing a product, whether it be a book, clothing, or electronic device, users first read opinions and evaluations of that product, which have an impressive impact on their decision-making (Hossain et al., 2021). SA produces an objective measure of fans' sentiment, thereby facilitating the evaluation of the overall success of a drama or film. Negative sentiment may indicate areas that require improvement, whereas

positive sentiment may indicate victorious audience reception (Sazzed, 2021). Furthermore, marketing research can employ the findings generated by SA techniques. This approach enables the examination of public opinion concerning a product, service, or recent governmental policy (Rambocas and Pacheco, 2018).

With the advancement of artificial intelligence (AI) and the evolution of machine learning (ML) models, plenty of research works focus on SA tasks in English and other highly-resourceful languages (Xu et al., 2022; Wankhade et al., 2022). However, we have detected few contributions in the low-resource Bengali language because of the standard annotated datasets and the computational resource limitations. Although, according to the report from the New World Encyclopedia (NWE), Bengali is the sixth most widely spoken language worldwide, with roughly 230 million native speakers. Moreover, Bengali is the official language of Bangladesh and the second most common language in India. Additionally, the 1952 Bengali language movement stimulated UNESCO to officially recognize February 21 as International Mother Language Day starting in 2000. Thus, we are inspired to experiment with the Bengali SA task in this research.

We find several research endeavors for classifying sentimental text in the Bengali language. Hasan et al. (2023) explored the subject of

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sentiment analysis in Ukrainian and Russian war news. They developed a dataset comprised of 10,861 entries. The BanglaBERT model attained the highest accuracy and F1-score at 86% and 0.82, respectively. [Bitto et al. \(2023\)](#) worked on another SA task related to food review. They collected just 1,400 texts in their experiment. The authors obtained the highest accuracy of 91.07% and F1-score of 85.28% by employing the Long Short-Term Memory (LSTM) and Extreme Gradient Boosting (XGBoost) models, respectively. In another research, [Haque et al. \(2023\)](#) conducted a study for classifying social media comments into political, religious, sexual, and acceptable categories. They applied a combined Convolutional Neural Network (CNN) and LSTM (CNN-LSTM) model with a word embedding technique for achieving the maximum output of 85.8% accuracy and 0.86 F1 score.

[Tareq et al. \(2023\)](#) collected a code-mixed Bangla-English corpus of 18,074 samples from the Google Play Store for SA. They showed XGBoost with fastText embedding performed a better result of 87% F1 score. On the other hand, [Bhowmik et al. \(2022\)](#) conducted SA on the ABSA cricket review dataset of 2,979 comments. They proposed a lexicon data dictionary (LDD) and a sentiment polarity extraction algorithm named Bangla text sentiment score (BTSC). By incorporating BERT-LSTM with LDD and BTSC, the authors obtained a better accuracy at 84.18%. [Hassan et al. \(2022\)](#) experimented with different contexts of SA with Bangla conversation data. They collected only 1,141 conversational texts from various Bangla short films and movie scripts. The authors got 85.59% accuracy by applying the Support Vector Machine (SVM) model.

[Protasha et al. \(2022\)](#) explored the matter of SA in online retailer product reviews from Facebook, Twitter, and YouTube comments. They developed a dataset consisting of 8,952 texts. They proposed a new approach by combining CNN and Bidirectional LSTM (BiLSTM) models with Bangla-BERT embedding, leading to an astounding accuracy of 94.15% and an F1-score of 93.04%. [Hossain et al. \(2021\)](#) analyzed Bengali book review sentiment data from 2,000 texts. They exploited multinomial Naive Bayes (MNB) with the unigram features method and got the highest accuracy of 87% and F1-score of 86%.

[Sazed \(2021\)](#) investigated Bengali drama review comments in the SA task. They developed a standard dataset containing 11,807 reviews. Authors utilized machine translation and combined lexicon-based and classical ML-based models. Their proposed hybrid-SVM method achieved the highest accuracy and F1-score of 91.5% and 0.89, respectively.

By investigating the recent works in Bengali SA, we see that most authors experimented on self-made small datasets. Furthermore, they rarely tried state-of-the-art transformer models which exhibit superior performance in various Bengali Natural Language Processing (NLP)-related downstream tasks, including SA, depressive text detection ([Hoque et al., 2024; Hoque and Salma, 2023](#)), cyberbullying text detection ([Hoque and Seddiqui, 2024a,b; Hoque et al., 2023](#)), and so on. Additionally, the authors placed little emphasis on methodological improvements for detecting sentiment in text. Moreover, sentiment text detection accuracy in Bengali is still far behind that of English and other high-resource languages. To overcome the above issues, we made the following contributions through this research:

- We utilize a recognized dataset of 11,807 comments in the Bengali SA task.
- We exploit five cutting-edge transformer models, including mBERT, Bangla-Bert-Base, BanglaBERT, DistilmBERT, and XLM-R-base by optimizing the hyperparameters.
- By combining transformer models, we suggest a new Transformer-ensemble method and investigate the rigorous qualitative analysis of this approach.
- Our proposed method shows remarkable performance over state-of-the-art works in classifying sentiment text for the low-resource Bengali language.

Table 1

Distribution of the number of comments in each sentiment class.

Sentiment label	Quantity	Percentage
Negative	3,307	28%
Positive	8,500	72%
Total	11,807	100%

2. Method

To determine the sentiment of a text, we first collect a publicly accessible dataset. Then we apply preprocessing tasks to eliminate the noisy contents. Subsequently, we employ transformer models by tuning the hyperparameters for extracting features and identifying the sentiment of a sample. We also utilize the ensemble approach in this experiment. Finally, we present a detailed comparative analysis of every pre-trained model. [Fig. 1](#) depicts the entire working procedure in this Bengali sentiment analysis task. We divide this section into five parts: dataset description, preprocessing tasks, feature selection tasks, transformer models, and experimental set-up. We illustrate these parts step-by-step as follows:

2.1. Dataset description

We use a well-annotated dataset of 11,807 samples of a comprehensive corpus of Bengali reviews ([Sazed, 2020](#)). The samples are collected from YouTube platforms, encompassing public opinions on Bengali dramas with positive and negative reviews. In this dataset, we use two columns: Text (containing public views) and Sentiment (Positive or Negative). [Table 1](#) articulates the number of reviews for each sentiment label.

2.2. Data preprocessing

The dataset conveys various noisy and unwanted elements. At first, we remove 1,041 duplicate entries from the dataset. Then, we filter URLs, HTML tags, special characters, punctuation, and digits from every sample in the dataset. After that, we discard other languages except Bengali. In this experiment, we do not consider stop-words removal and stemming operations for getting maximum variations of features ([Kumar et al., 2021](#)).

2.3. Feature selection

Every transformer-based pre-trained model exploits the self tokenization method for extracting sub-word level tokens (feature vectors). The mBERT, BanglaBERT, Bangla-Bert-Base, and DistilmBERT utilize the WordPiece technique, whereas the XLM-R employs Sentence Piece Model (SPM) ([Kudo and Richardson, 2018](#)) approach. The SMP combines two sub-word segmentation techniques, such as byte-pair-encoding (BPE) ([Sennrich et al., 2015](#)) and uni-gram language model ([Kudo, 2018](#)).

2.4. Transformer models

We utilize five state-of-the-art transformer models, such as mBERT, BanglaBERT, Bangla-Bert-Base, DistilmBERT, and XLM-R-base for the Bengali sentiment analysis task. These models show outstanding performance in sentiment analysis and other Bengali (NLP) tasks ([Hoque et al., 2024; Islam et al., 2020; Hasan et al., 2023; Saha et al., 2022; Hoque et al., 2023](#)). The mBERT ([Pires et al., 2019](#)) operates according to the same principle as the original BERT ([Devlin et al., 2018](#)) model. The mBERT model pre-trained over 104 languages. It uses pre-training and fine-tuning frameworks that have multi-head self-attention layers. Every self-attention is calculated by Eq. (1), and multi-head attention

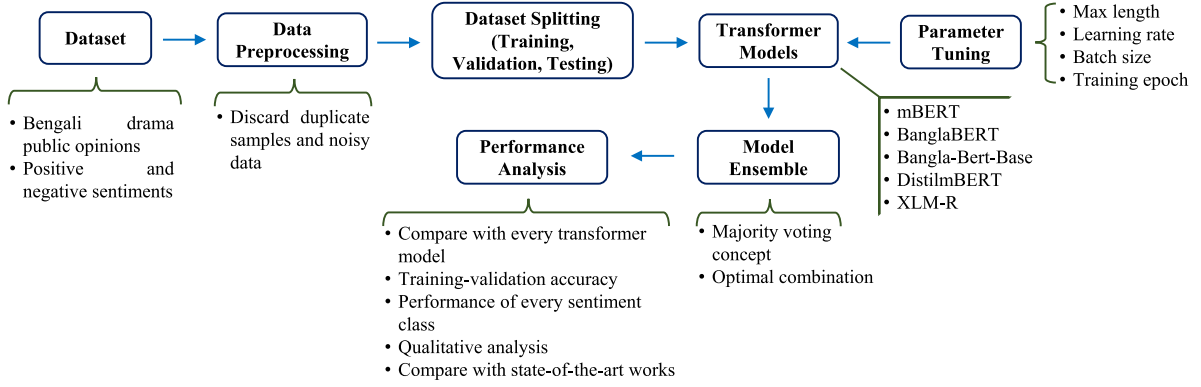


Fig. 1. Overall working process.

is calculated using Eqs. (2) and (3) (Vaswani et al., 2017). In the pre-training framework, the mBERT employs two tasks: the Mask Language Model (MLM) and Next Sentence Prediction (NSP), for a comprehensive understanding of a specific language. On the other hand, the fine-tuning framework provides various NLP-related downstream tasks, such as sentiment classification and others, through simple modifications to the task-specific architecture. The BanglaBERT (Bhattacharjee et al., 2021) is pre-trained using more than 27.5 GB of data extracted from 110 renowned Bengali websites. The model comes by pre-training the ELECTRA (Clark et al., 2020) model. It has two principal components: the generator and the discriminator. The generator utilizes the MLM like BERT for masking 15% of the entire tokens. Then it replaces these masked tokens with equivalent terms. The discriminator part exploits the Replaced Token Detection (RTD) objective instead of BERT's NSP objective to specify whether or not every single token comes from the original text sequence. Another Bengali-specific pre-trained model is Bangla-Bert-Base (Sarker, 2020). It utilizes two Bengali corpora: Bengali commoncrawl corpus gathered from OSCAR and Bengali Wikipedia Dump Dataset. It derives from pre-training the BERT (Devlin et al., 2018) model and follows the standard MLM objective of BERT. The DistilmBERT is a fast, cheap, and lightweight transformer model that follows the mBERT model's architecture. Knowledge distillation during the pre-training stage reduces time and space significantly compared to the mBERT model. The XLM-R-base (Conneau et al., 2019) is a cross-lingual transformer model pre-trained on over 2 TB (terabytes) of data. It shows a better performance, particularly for the low-resource languages. Table 2 articulates the model architecture of the five transformer models, where L is the transformer layers, E is the embedding size, H is the hidden size, H_{ff} is the hidden intermediate feed-forward layer size, A is the attention head, V is the vocabulary size, $G2D$ generator-to-discriminator ratio and P is the number of parameters.

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \quad (1)$$

where, Q , K , and V mention the query, key, and value matrices, respectively, d_k indicates the dimension of Q and K , and $\frac{1}{\sqrt{d_k}}$ represents the scaling factor.

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \quad (2)$$

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (3)$$

where, h indicates the number of attention head, W^O is the output weight of the attention unit, and W_i^Q , W_i^K , and W_i^V present the i th attention weight of Q , K , and V matrices, respectively.

Ensemble of Transformer models: The aim of the model ensemble technique is to enhance system performance by combining multiple models. In this research, we ensemble the transformer models by following the strategy outlined in the Algorithm 1. At first, we input the predicted class label of each transform model m_i ($i = 1, 2, \dots, n$),

Table 2

Architecture of the transformer models.

Models	L	E	H	H_{ff}	A	V	$G2D$	P
mBERT	12	768	768	3072	12	110 k	–	172 M
BanglaBERT	12	768	768	3072	12	32 k	$\frac{1}{3}$	110 M
Bangla-Bert-Base	12	768	768	3072	12	102 k	–	110 M
DistilmBERT	6	768	768	3072	12	120 k	–	134 M
XLm-R-base	12	768	768	3072	12	250 k	–	270 M

and the size of the test sample n_{ts} , where n is the number of models. The inner For loop (lines 5 to 8) finds the predicted class label of every model for a particular instance. Finally, line 9 calculates the highest-voted class label and stores this label in the ensemble model variable m_{ens} . Throughout the empirical experiment, we see the ensemble of the top three models: BanglaBERT (m_1), XLM-R-base (m_2), and mBERT (m_3), outperforms any combination of the five pre-trained models.

Algorithm 1: Transformer-ensemble method

Input: $m_1, m_2, \dots, m_n$: Predicted class label of n trained models
 n_{ts} : Total number of test samples
Output: m_{ens} : Predicted class label of ensemble method

```

1  $m_{ens} \leftarrow []$  /* Empty list */
2  $m_{temp} \leftarrow [0] \times n$  /* Initialize a list with  $n$  number of zeros */
3  $i \leftarrow 0$ 
4  $j \leftarrow 1$ 
5 for  $i \leq n_{ts} - 1$  do
6   for  $j \leq n$  do
7      $m_{temp}[j-1] \leftarrow m_j[i]$  /* Find the predicted class label of every model for each sample */
8      $j \leftarrow j + 1$ 
9    $m_{ens} \leftarrow append(majorityVoting(m_{temp}))$  /* Find the highest-voted class label and store it in the ensemble variable */
10   $i \leftarrow i + 1$ 

```

2.5. Experimental set-up

In this investigation, we use a dataset consisting of 11,807 samples. Additionally, we employ transformer-based pre-trained models. For these reasons, we require hardware with high configurations to implement transformer models on the dataset. Thus, we utilize a cloud Google Colab platform that facilitates a Jupyter Notebook editor with many Python modules and packages. The Colab provides a powerful Tesla T4 NVIDIA Graphics Processing Unit (GPU) of 15 GB RAM.

While incorporating pre-trained models, we partition our dataset into training, validation, and testing subsets, upholding a uniform distribution of 70%, 10%, and 20%, respectively. We exploit the

Table 3

Overview of the hyperparameters.

Parameter	Data type	Description	Value
Maximum token length (<i>tkLen</i>)	Integer	It indicates the maximum number of tokens per review comment.	Min value = 100 and Max value = 192
Optimizer (AdamW) learning rate (<i>lr</i>)	Float	It modifies the learning rate where a loss function proceeds to the point in which the curves converge.	Value=[1e-05, 2e-05, 3e-05, 4e-05, 5e-05, 6e-05]
Training epochs (<i>epoch</i>)	Integer	It refers to the number of passes of the complete training data to train the model.	Min value = 2 and Max value = 15
Training batch size (<i>bSize</i>)	Integer	It denotes the number of samples passed in every iteration of each epoch during model training.	Min value=12 and Max value=40

Table 4

The best model configuration.

Models	<i>tkLen</i>	<i>lr</i>	<i>epoch</i>	<i>bSize</i>
mBERT	128	5e-05	10	12
BanglaBERT	128	5e-05	10	12
Bangla-Bert-Base	132	4e-05	10	12
DistilmBERT	128	5e-05	10	12
XML-R-base	132	4e-05	10	12

Ktrain python module for applying the five models, including mBERT, BanglaBERT, Bangla-Bert-Base, DistilmBERT, and XML-R-base. This research investigates four performance-changing parameters to get the best optimal setting of each transformer model. Table 3 specifies these parameters by indicating their range of values. We decide the minimum and maximum values throughout the empirical investigations and studying of the other research works (Hoque and Seddiqui, 2024b, 2023; Hoque and Salma, 2023) relating to the encoder-based transformer models. We perform rigorous experiments to fix the best possible combination of the four parameters, as shown in Table 4.

3. Results and discussion

To calculate the performance of every transformer model, we employ four evaluation metrics, including precision, recall, F1-score, and accuracy. We can measure these metrics based on the four key terms: (a) True Positive (TP) — accurately predicted a total amount of instances as a positive class, (b) True Negative (TN) — correctly predicted a number of samples as a negative class, (c) False Positive (FP) — incorrectly predicted a number of negative class samples as a positive class, and (d) False Negative (FN) — incorrectly predicted a number of positive class samples as a negative class. The paper uses the following equations to compute the four evaluation metrics:

$$\text{Precision (P)} = \frac{TP}{TP + FP}$$

$$\text{Recall (R)} = \frac{TP}{TP + FN}$$

$$\text{F1-score (F)} = 2 \times \frac{P \times R}{P + R}$$

$$\text{Accuracy (A)} = \frac{TP + TN}{TP + TN + FP + FN}$$

Since we work on an imbalanced dataset, the weighted average scores consider in this investigation. The class-wise weight value is calculated as follows:

$$wtCls^i = \frac{nObs^i}{\sum_{j=0}^{nCls-1} nObs^j}$$

where, *wtCls* denotes the weight value of *i*th class, *nObs* indicates the amount of supported observations of a particular class, and *nCls* represents the number of classes. Finally, the paper calculates the weighted average score for each performance metric (except accuracy) as follows:

$$wtAvg = \sum_{i=0}^{nCls-1} metricCls^i \times wtCls^i$$

Table 5

Performance comparison of various transformer models.

Model	Precision	Recall	F1-score	Accuracy
mBERT	94.16%	94.20%	94.17%	94.20%
BanglaBERT	95.81%	95.68%	95.72%	95.68%
Bangla-Bert-Base	93.37%	93.41%	93.39%	93.41%
DistilmBERT	93.62%	93.64%	93.63%	93.64%
XML-R-base	95.22%	95.22%	95.22%	95.22%
Transformer-ensemble	95.96%	95.98%	95.96%	95.97%

Here, *wtAvg* specifies a weighted average value for a particular evaluation metric, and *metricCls* denotes the class-wise score of a performance metric. We divide this section into two parts. The first part describes the output results and comparative performance of every transformer model, and the second part illustrates a detailed discussion about the achievement of each transformer model and our proposed method. Additionally, it includes a comparative analysis with existing methods.

3.1. Result analysis

In Table 5, we articulate the results of every transformer model. Among the five individual models, the BanglaBERT achieves the highest outputs of 95.81% precision, 95.68% recall, 95.72% F1-score, and 95.68% accuracy. The XML-R also shows better outputs like BanglaBERT and obtains the second-highest position in all four evaluation metrics. The performance ranking of mBERT is third, DistilmBERT is fourth, and Bangla-Bert-Base is fifth. According to the Algorithm 1, our proposed Transformer-ensemble method outperforms every individual model with precision, recall, F1-score, and accuracy of 95.96%, 95.98%, 95.96%, and 95.97%, respectively. Here, we consider the top three models, such as BanglaBERT, XML-R-base, and mBERT, to obtain the best outputs.

3.2. Detailed discussion

Since our proposed Transformer-ensemble composes BanglaBERT, XML-R-base, and mBERT models, we begin by illustrating the training and validation accuracy curve of these three models. Then, we explore the evaluation results of each sentiment class of every transformer model. Subsequently, we discuss the qualitative analysis of our proposed method. Finally, we compare the performance of this method with the recent approaches for classifying Bengali sentiment text.

Analysis of training-validation accuracy curve: Fig. 2 depicts the training-validation accuracy curves of mBERT, BanglaBERT, and XML-R-base of 10 epochs. In Fig. 2(a), the training accuracy of the mBERT model shows about 0.87 value in the first epoch. The curve gradually rises from the third epoch (about 0.94) to the last epoch (close to 1.0). On the contrary, the validation curve presents slight ups and downs with an upward trend, where in the first and the last epochs, the validation accuracy is near about 0.90 and 0.94, respectively. In Fig. 2(b), the training accuracy curve of the BanglaBERT model displays a similar pattern to the mBERT model, while the accuracy score starts at 0.90 and ends with close to 1.0. Conversely, the validation accuracy begins with a value of around 0.95 (first epoch) and reaches nearly

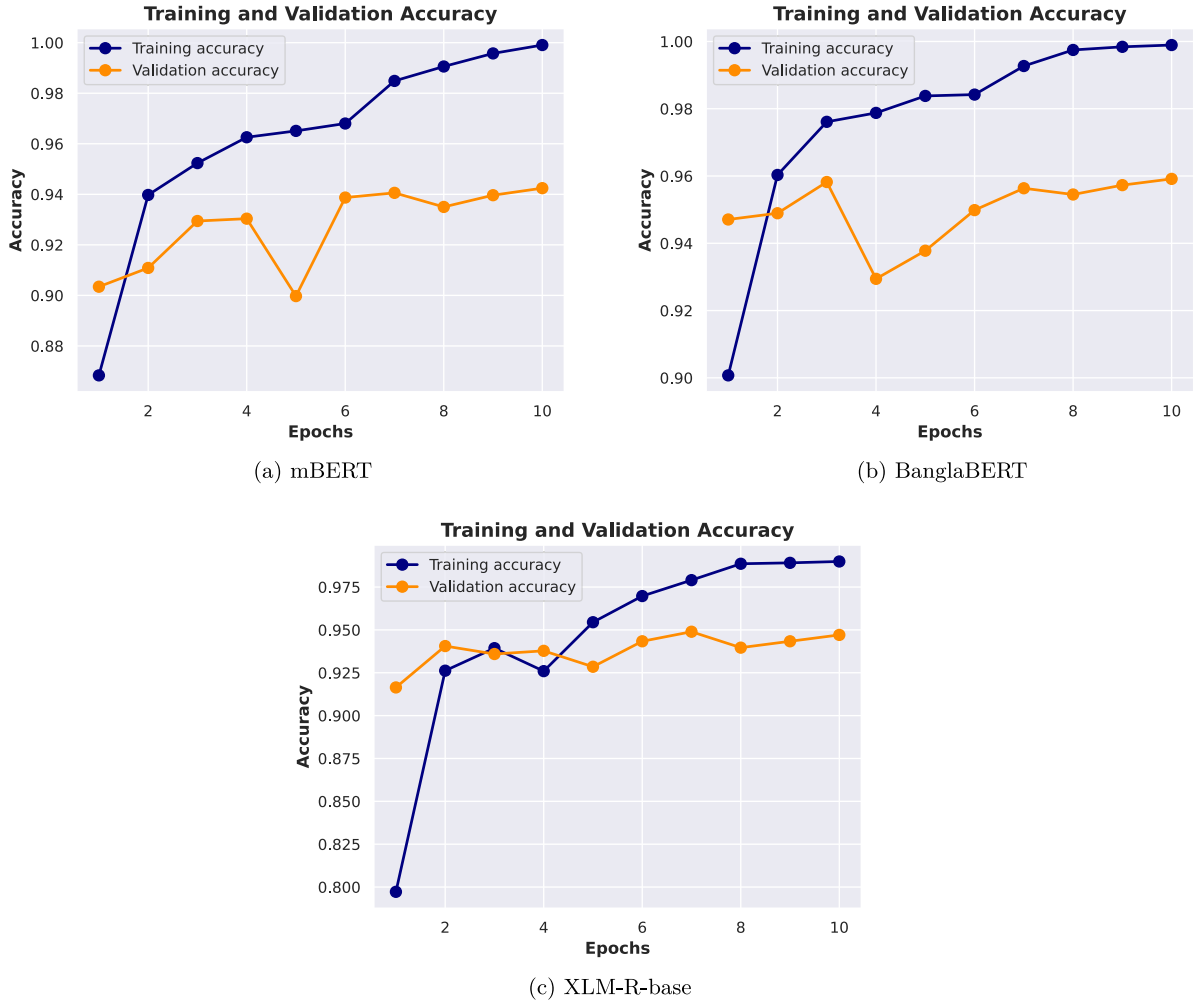


Fig. 2. Training-validation accuracy curves of the transformer models.

0.96 by the tenth epoch. In between the first and the last epochs, the validation accuracy fluctuates at some points. The training accuracy of the XLM-R-base model in Fig. 2(c) unstably rises near from 0.80 to 0.93 for the first to the fourth epoch. After that curve gradually increases up to the final epoch by reaching an accuracy score of approximately 0.99. On the other side, the validation accuracy curve visualizes an ups and downs pattern with a slight upward direction. Here, the starting accuracy is about 0.93, and the final value is close to 0.95.

Class-wise performance analysis: We delineate the performance of every model for the Negative and Positive classes sentiment through Table 6 and the confusion matrix diagrams in Fig. 3. We observe that our proposed Transformer-ensemble method achieves the best F1 scores of 92.94% and 97.17% for the Negative and Positive classes, respectively. This method also shows the highest precision value (92.42%) for the Negative class and the recall value (96.95%) for the Positive class. In Fig. 3, we see that the FN value of the Negative class, is high for every model except the BanglaBERT. Since a lower FN value increases the recall rate, the BanglaBERT shows a maximum recall of 95.27% for this class. Similarly, the lower FP value of this model gives a top precision score of 98.07% for the Positive class. Among the five individual models, the mBERT and XLM-R-base obtain the same higher recall value of 96.66% for the Positive class because of a lower FP value. The Bangla-Bert-Base and DistilmBERT display lower performance for both classes in every evaluation metric.

Qualitative analysis of the proposed method: Our proposed Transformer-ensemble method combines three pre-trained models: mBERT, BanglaBERT, and XLM-R-base. It is highly biased by the

Table 6
Class-wise performance evaluation of the transformer models.

Model	Class	Precision	Recall	F1-score
mBERT	Negative	91.22%	88.09%	89.63%
	Positive	95.33%	96.63%	95.97%
BanglaBERT	Negative	90.12%	95.27%	92.62%
	Positive	98.07%	95.85%	96.95%
Bangla-Bert-Base	Negative	89.18%	87.44%	88.30%
	Positive	95.04%	95.78%	95.41%
DistilmBERT	Negative	89.40%	88.90%	88.74%
	Positive	95.29%	95.85%	95.57%
XLM-R-base	Negative	91.53%	91.68%	91.61%
	Positive	96.69%	96.63%	96.66%
Transformer-ensemble	Negative	92.42%	93.47%	92.94%
	Positive	97.39%	96.95%	97.17%

BanglaBERT model, a Bengali language-specific pre-trained model. Additionally, its model architecture (embedding size and discriminator part) is slightly different than the other transformer models (see Table 2). On the other hand, the mBERT and XLM-R-base have a large vocabulary and a large number of parameters than the BanglaBERT model. Because of these, our proposed method gives superior performance than the five individual models in four evaluation metrics (see Table 5). We also visualize some samples in Table 7 (the Negative class is labeled by 0 and the Positive is 1), where the proposed method gives true prediction than the other models. In addition, the research examines Cohen's Kappa scores (Cohen, 1960) with respect to the actual labeling of the test data for every approach. A higher Kappa

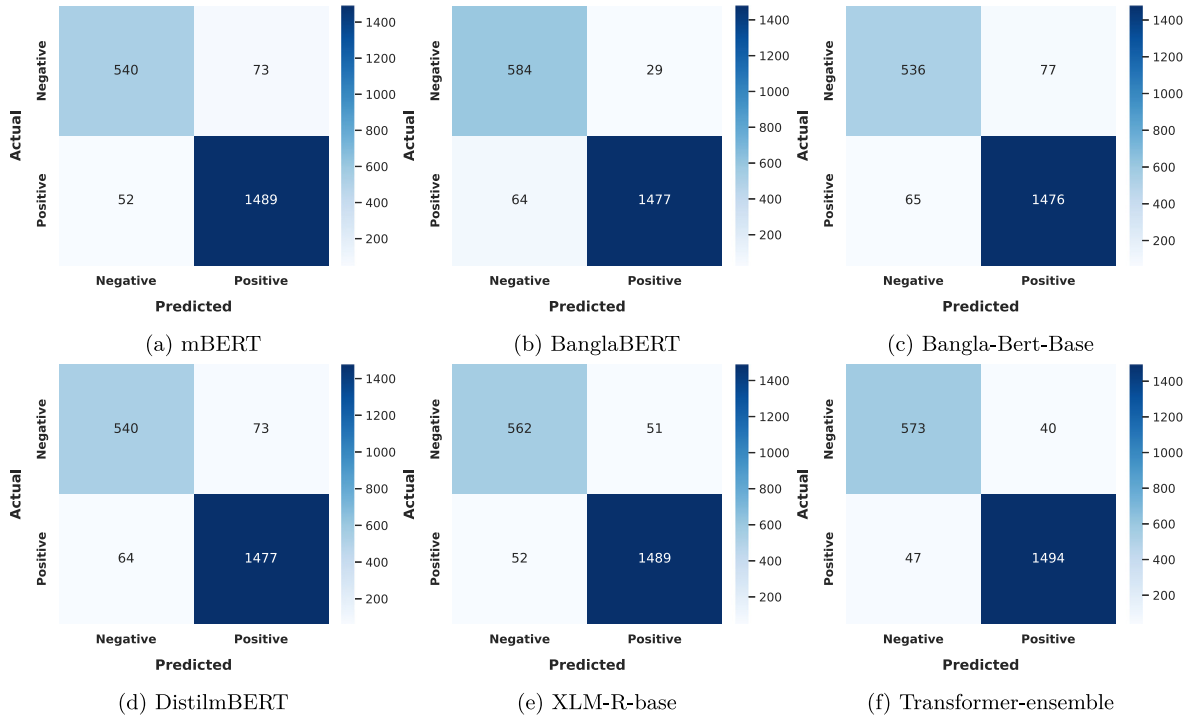


Fig. 3. Confusion matrix of the transformer models.

Table 7

Justifying the quality of the Transformer-ensemble classifier with the other models.

Sentiment text	Predicted labels						Gold label
	mBERT	Bangla-BERT	Bangla-Bert-Base	Distil-mBERT	XLM-R-base	Transformer-ensemble	
ফিনিশিং সুন্দর হয়নি, আরো নাটকীয় হওয়ার দরকার ছিলো। (The finishing was not pretty, needed to be more dramatic.)	0	1	0	1	0	0	0
Thankyou ক্ষনিকের আনন্দ দেওয়ার জন্য (Thank you for the momentary pleasure)	0	1	0	0	1	1	1
টাকা থাকলে সবাই sorry বলে... (Everyone says sorry if they have money...)	0	0	1	1	1	0	0

Table 8

Cohen's Kappa scores of the transformer-based approaches.

Approaches	Cohen's Kappa scores
mBERT	0.8560
BanglaBERT	0.8958
Bangla-Bert-Base	0.8371
DistilBERT	0.8431
XLM-R-base	0.8826
Transformer-ensemble (Proposed)	0.9011

value signifies a greater degree of agreement in labeling between the predicted labels and the original labels of the test data, indicating higher reliability. According to Table 8, the proposed technique achieves a higher value of 0.9011 compared to the other approaches. This number enhances the dependability of the suggested detection system.

The Transformer-ensemble method presents lower performance for the Negative class compared to the Positive. The main reason is the percentage of the Negative sentiment data is very low (28%) than the positive one (72%), which deteriorates the performance for recognizing the Negative class data. Moreover, some samples in our dataset contain negative words, although contextually these are positive. For example,

in a text “নাটক টা যে বানাইছে শালা একটা মাল। ভালো লাগছে।” (The drama that is being made by a person is a genius. Feeling good.), the words “শালা” (shala) and “মাল” (mal) are commonly used as a negative sense. However, the contextual meaning of these words is positive in this text. Our detection system misclassifies this type of sample. Again, since we discard all other languages except Bengali, sometimes the non-Bengali words in a sample determine the sentiment of an entire text. For example, a non-Bengali word “faltu” in an instance “লাইফে fast time এমন faltu নাটক দেখলাম...” (For the first time in my life, I saw such a bogus drama...), determines the negative sentiment for this text. Our method cannot handle this kind of sample. Furthermore, since the dataset consists of unstructured data, misspelled words, various regional words, and others, our proposed method faces difficulty in recognizing Bengali sentiment classes.

Compare with related work: In the Bengali SA task, the authors studied on different context of sentiment, such as Russia-Ukraine War, food review, and others, utilized distinct-sized of dataset with various classes of sentiment, and exploited variety of approaches (see Table 9). In this table, we also notify the performance of each classification system. We observe that our proposed Transformer-ensemble method shows outstanding results than the every existing approach with F1-score of 95.96% and accuracy of 95.97%.

Table 9

Performance comparison with the related works of the Bengali Sentiment Analysis.

Approach	Context	Dataset size	Sentiment classes	F1-score	Accuracy
Bangla BERT (Hasan et al., 2023)	Russia–Ukraine War	10,861	3 Classes: Pro-Ukraine, Pro-Russia, and Neutral	82.00%	86%
XGB, LSTM (Bitto et al., 2023)	Food review	1,400	2 Classes: Positive and Negative	85.28% (XGB) 83.22% (LSTM)	89.64% (XGB) 91.07% (LSTM)
CNN–LSTM (Haque et al., 2023)	Social media comments	42,036	4 classes: Acceptable, Political, Religious, and Sexual	86.00%	85.80%
XGBoost with cm-FastText (Tareq et al., 2023)	Google Play Store comments	18,074	5 classes: Positive, Negative, Mixed positive, Mixed negative and Neutral	87.00%	–
BERT-LSTM with LDD and BTSC (Bhowmik et al., 2022)	ABSA Cricket dataset	2,979	3 classes: Positive, Negative, and Neutral	82.27%	84.18%
SVM (Hassan et al., 2022)	Movies and short films scripts conversation	1,141	2 classes: Positive and Negative	86.64%	85.59%
CNN–BiLSTM with Bangla-BERT (Prottasha et al., 2022)	Internet and social media sites	8,952	2 classes: Positive and Negative	93.04%	94.15%
MNB with unigram (Hossain et al., 2021)	Book review	2,000	2 classes: Positive and Negative	86.00%	87.00%
Self-Supervised-Hybrid-SVM (Sazzed, 2021)	Drama review	11,807	2 classes: Positive and Negative	89.70%	91.50%
Transformer-ensemble (Proposed)	Drama review	11,807	2 Classes: Positive and Negative	95.96%	95.97%

4. Conclusion

For current researchers, successfully extracting the sentiment from a Bengali text is more challenging. In this regard, we employ five cutting-edge transformer-based pre-trained models and develop a new Transformer-ensemble method. This method combines three pre-trained models: mBERT, BanglaBERT, and XLM-R-base, and outperforms the existing detection systems with excellent accuracy of 95.97% and F1-score of 95.96%. Additionally, we investigate a critical qualitative analysis of our proposed method. Our sentiment detection system aids the general people to know the quality of the product, such as movies, dramas, and others. Besides, the authority of the product may be more conscious of fulfilling the demand of the users by using this system. In addition, our proposed method may help in other decision-making review systems like food review. However, this system sometimes shows slightly below performance for the Bengali–English code-mix data. Besides, because of the low percentage of Negative class data, this method struggles to identify samples belonging to this class. In the future, we will handle the problem of negative class samples and other issues. Additionally, we will experiment on a large dataset with more sentiment classes and propose another new and robust hybrid-model approach for the sentiment analysis task. Finally, we will develop a real-time mobile and web-based application system for detecting the sentiments in a text more accurately.

CRedit authorship contribution statement

Md. Nesarul Hoque: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Umme Salma:** Validation, Writing – original draft, Writing – review & editing. **Md. Jamal Uddin:** Validation, Writing – original draft, Writing – review & editing. **Md. Martuza Ahamad:** Validation, Writing – original draft, Writing – review & editing. **Sakifa Aktar:** Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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